

INDUSTRIAL INTERNSHIP REPORT

BANKING INFORMATION SYSTEM

Python Internship Project

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Item	Details
Project Domain	Python
Project Type	Console-based Banking Application
GitHub Username	vickey-raj-cs28
Repository	https://github.com/vickey-raj-cs28/upskillcampus
Code Link	https://github.com/vickey-raj-cs28/upskillcampus/blob/main/banking_information_system.py
Report Link	https://github.com/vickey-raj-cs28/upskillcampus/blob/main/Vickey_BankingInformationSystem_Final_Report.pdf

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1 Preface

This industrial internship project was completed to gain practical experience in Python programming and project-oriented problem solving. The final project is a console-based Banking Information System that combines fundamental Python concepts into a practical application.

The application accepts user input, stores account records, updates balances, records transactions and persists information in a JSON file. It demonstrates functions, lists and dictionaries, `append()`, `input()`, loops, conditions and file handling.

The report follows the organization of the supplied internship sample, including project description, proposed solution, design/model, testing, learnings and future scope.

2 Introduction

2.1 About the Internship

The supplied sample report presents the internship as a structured practical learning and project-development experience in which learners work on an assigned problem statement and document implementation, testing and learning outcomes.

2.2 Objectives

- Develop practical Python programming skills.
- Apply programming concepts to a complete project.
- Improve logical thinking and problem-solving ability.
- Practice debugging, validation and testing.
- Document and submit a project professionally.

2.3 Glossary

Term	Meaning
Python	Programming language used for the project.
List	Collection used to store multiple values.
append()	Method used to add an item to a list.
Function	Reusable block of code defined with def.
File handling	Reading and writing persistent information.
JSON	Text format used to store account data.
GitHub	Platform used to store and share source code and reports.

3 Problem Statement

Basic banking operations require maintaining customer information, account balances and transaction records. This project implements a small educational Banking Information System that allows a user to create an account and perform basic account operations while keeping the information available between program runs.

Functional requirements:

- Create an account with customer name, account number and opening balance.
- Prevent duplicate account numbers.
- Deposit money into an existing account.
- Withdraw money when sufficient balance is available.
- Display current balance.
- Display transaction history.
- Store account information in a file.

Scope: This is an educational application and is not intended for real banking, authentication, encryption, regulatory compliance or handling real financial information.

4 Existing and Proposed Solution

4.1 Existing Solution

A manual approach requires account information and transaction records to be maintained and updated separately. This can be repetitive and makes consistent validation more difficult.

4.2 Proposed Solution

The proposed solution is a menu-driven Python application. Separate functions handle account creation, deposits, withdrawals, balance checking and transaction history. Structured account records are stored in memory and persisted in a JSON file.

- User interaction through input().
- Reusable functions using def.
- Lists and dictionaries for structured records.
- append() for transaction records.
- JSON file persistence.
- Validation for invalid input and insufficient balance.

4.3 GitHub Submission Links

Repository	https://github.com/vickey-raj-cs28/upskillcampus
Code	https://github.com/vickey-raj-cs28/upskillcampus/blob/main/banking_information_system.py
Final Report	https://github.com/vickey-raj-cs28/upskillcampus/blob/main/Vickey_BankingInformationSystem_Final_Report.pdf

The report URL is the final location of this PDF in the GitHub repository. The PDF should be uploaded at this exact filename so that the link remains valid.

5 Proposed Design / Model

5.1 High Level Flow

Step	Process
1	Start application
2	Load saved account data from accounts.json
3	Display main menu
4	Accept user's choice
5	Call the selected function
6	Validate input and update data
7	Save changes to accounts.json
8	Display result and return to menu
9	Exit when option 6 is selected

5.2 Low Level Flow

- `load_accounts()` reads the JSON file.
- `find_account()` searches for an account number.
- `create_account()` collects details and uses `append()` to add a new account.
- `deposit_money()` increases balance and records a transaction.
- `withdraw_money()` validates amount and available balance.
- `check_balance()` displays account information.
- `show_transactions()` displays transaction records.
- `save_accounts()` writes updated data to `accounts.json`.

5.3 Interfaces

The project uses a command-line interface. Users interact through numbered menu options and text prompts. `accounts.json` provides persistent storage.

6 Performance Test

The project was tested through functional cases covering normal operations and common invalid inputs.

6.1 Test Cases

ID	Operation	Condition	Expected Result
TC-01	Create account	Valid unique details	Account created
TC-02	Duplicate account	Existing account number	Creation rejected
TC-03	Deposit	Positive amount	Balance increases
TC-04	Withdraw	Amount within balance	Balance decreases
TC-05	Insufficient balance	Amount exceeds balance	Withdrawal rejected
TC-06	Invalid amount	Non-numeric input	Error handled
TC-07	Balance	Existing account	Correct balance displayed
TC-08	History	Existing account	Transactions displayed
TC-09	Persistence	Restart after save	Saved accounts loaded

6.2 Test Procedure

- Run the program using Python 3.
- Execute each menu operation with valid and invalid inputs.
- Compare output with expected behavior.
- Check that account data is written to and read from accounts.json.
- Repeat operations after restarting the program.

6.3 Performance Outcome

The implementation is intended for small educational datasets and provides immediate console responses for individual operations. A production banking system would require a database, authentication, encryption, auditing, concurrency controls and stronger security.

7 My Learnings

- Improved understanding of Python syntax and program structure.
- Practiced input() for user interaction.
- Used lists and dictionaries for data organization.
- Used append() to add accounts and transactions.
- Created reusable functions using def.
- Learned file handling and JSON persistence.
- Practiced validation and exception handling.
- Improved debugging and logical problem solving.
- Learned to organize a project for GitHub submission.

8 Future Work Scope

- Add login and role-based authentication.
- Replace JSON storage with a database.
- Add account update and deletion features.
- Add fund transfer functionality.
- Improve validation and error logging.
- Create a graphical or web interface.
- Add automated unit tests.
- Add secure credential and data handling for production-oriented use.
- Deploy the application if required.

Final Submission Checklist

- Repository name: upskillcampus.
- Upload banking_information_system.py.
- Upload accounts.json, README.md and requirements.txt.
- Upload Vickey_BankingInformationSystem_Final_Report.pdf.
- Verify the code and report URLs.
- Keep the final PDF filename exactly as shown in the report link.
- Test the repository and links before submission.